

Using AI for Visualization

Big Data Summer Immersion · BDSY 2026

Friday, June 26 · 10:45 AM · Room 106A

Last morning of Week 2 — we tie your Python + ggplot2 week together

Why this matters (it's not just pretty pictures)

- A chart is how data **makes an argument** — in public health, to clinicians, funders, the public.
- A **misleading** chart doesn't just look bad — it **misinforms**.
- AI can now draft a chart in seconds. That makes **your judgment** the scarce skill, not the typing.

The one rule today: **AI drafts, you direct.**

seaborn = ggplot2's Python cousin

You already know the idea: **map data columns to visual properties**.

ggplot2 (yesterday)	seaborn (today)
<code>ggplot(df, aes(x, y, color=g))</code>	<code>sns.scatterplot(data=df, x=, y=, hue="g")</code>
<code>color =</code> / <code>fill =</code>	<code>hue=</code>
<code>facet_wrap(~group)</code>	<code>col="group"</code>
<code>geom_point()</code> / <code>geom_bar()</code>	<code>scatterplot()</code> / <code>barplot()</code>

Same grammar of graphics. Python syntax. Good defaults out of the box.

The loop you already met in *Coding with Claude*

describe → generate → run → **read** → refine

- Works with **any** assistant: ChatGPT, Claude, Copilot — or **Gemini, built into Colab** (nothing to install).
- The skill is the **loop and the critical eye**, not the brand.
- Reading the **error** (or the output) is where the learning happens.

Anatomy of a good viz prompt

1. **Dataset & columns** → name the DataFrame and exact columns
2. **Goal** → the question / the takeaway
3. **Constraints** → "use seaborn", colorblind-safe, one figure, clear title
4. **Chart type** (*optional*) → if you already know it, say so

```
Using the DataFrame `penguins` (columns: species, body_mass_g, sex, ...),  
use seaborn to plot body_mass_g per species as a boxplot, colored by sex,  
colorblind-safe palette, clear title. Return only code.
```

The dangerous failures are the *plausible* ones

- AI rarely fails with obvious gibberish.
- It borrows a feature from **another library** and writes code that *looks* right — e.g. `trendline="ols"` (that's **Plotly**, not seaborn) → it errors.
- Worse: code that **runs but quietly plots the wrong thing**.

So we always **verify before we trust**.

Trust-this-figure checklist

Before you believe **any** AI-generated plot:

- [] Uses **real** seaborn functions/arguments (nothing borrowed from another library)
- [] **Chart type matches the question** (right family)
- [] **Axes, units, title** are correct and honest
- [] **Numbers match the data** — spot-check one value
- [] A stranger could read it **without you explaining it**

AI drafts. You direct. You verify.